UNIVERSITY OF INFORMATION TECHNOLOGY AND SCIENCES (UITS)
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CSE 416: Software Project Design and Development (Lab)

SOFTWARE TEST REPORT

The Round Table Exchange (RTE): A Multi-Party Skill Barter Network with Graph-Based Cycle Matching

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Document Control

Parameter	Configuration Reference
Repository	https://github.com/Veil-Proxima/RTE
Version Tested	v1.0.1-test (commit 141fe65)
Live Client Site	https://veil-proxima.github.io/RTE (Local API: http://127.0.0.1:8000)
Report Version	1.0 (Baseline quality tier)
Prepared By	Ahammed Abdali Khan, Sumaia Bintey Ismail, Alin
Reviewed By	Ahammed Abdali Khan
Target Quality Tier	Baseline (ISO/IEC/IEEE 29119-3 & ISO/IEC 25010 compliant)

Document Revision History

Version	Date	Summary of Verification Changes
1.0	02 Nov 2026	**First comprehensive baseline report: 78 automated test cases, unit + integration suites, RTM traceability, 98% core backend coverage, NFR performance benchmarks, and defect log.**
1.1	05 Nov 2026	Planned for Exp 10: Extended OWASP ZAP automated scanner, SonarCloud static analysis gates, and multi-browser Playwright matrix.

1. Introduction & Overview

1.1 Purpose: This report details the verification of RoundTable Exchange (RTE) v1.0.1-test: test scope, methodology, automated test results, code coverage telemetry, defect analysis, and release readiness based on ISO/IEC/IEEE 29119-3 and ISO/IEC 25010.

1.2 Product Overview: RTE is a circular multi-party skill barter engine that eliminates the classic 'Double Coincidence of Wants' barrier by discovering closed barter cycles of length k in {2, 3, 4} using a Bounded DFS multigraph engine. Discovered cycles are evaluated via a composite function balancing Haversine geographic proximity with weekly availability schedules: S(C) = alpha * S_geo(C) + (1 - alpha) * S_sched(C). An atomic state machine coordinates unanimous single-veto consensus.

1.3 Scope of Testing

Subsystem / Functional Area	SRS Requirements	Testing Levels
User onboarding & Dhaka coordinate profiles	FR-01 to FR-05, FR-24	Unit, Integration
Skill taxonomy & multi-listing management	FR-06 to FR-10	Unit, Integration
Directed Multigraph construction & edges	FR-11 to FR-14	Unit, Integration
Bounded DFS cycle discovery (k in [2, 4])	FR-15 to FR-18	Unit, Integration, Algorithmic
Rotational cycle deduplication & DAG safety	FR-19 to FR-21	Unit, Algorithmic
Haversine geodesic distance ranking	FR-22 to FR-23	Unit, Integration
Weekly availability grid set intersection	FR-24 to FR-26	Unit, Integration
Dynamic composite scoring & real-time alpha	FR-27 to FR-28	Unit, Client Engine
Atomic 5-stage proposal state machine	FR-29 to FR-33	Unit, Integration
Synthetic population generator & scaling	FR-34 to FR-36	Performance, Load
REST API gateway contract validation	FR-37 to FR-40	Integration, Security
Client-side standalone JavaScript engine	FR-41 to FR-43	Client Unit, Parity
Responsive ForceAtlas2 physics canvas	FR-44 to FR-46	UI, Cross-Browser
Performance, Security, and Code Coverage	NFR-01 to NFR-16	Perf, Sec, Coverage

2. Test Environment & Strategy

Component	Specification & Tooling Configuration
Host Operating System	Windows 11 Enterprise (local test harness); Linux x86_64 (CI container)
Language Runtimes	Python 3.15.0 (64-bit AMD64), Node.js v24.22.1
Backend Frameworks	FastAPI 0.115, Starlette, Pydantic v2.10, Uvicorn 0.34, NetworkX 3.4
Test Runners & Libraries	pytest 9.1.1, pytest-cov 6.0, coverage.py 7.6, anyio 4.15, Node test runner
Test Store	In-memory Multigraph and clean dictionary store; fresh instances on every test run
Browsers Tested	Google Chrome 131, Microsoft Edge 131 desktop, responsive mobile viewports

2.1 Exit Criteria Evaluation (Baseline Quality Tier)

Evaluation Criterion	Target Metric	Achieved Result	Status
Functional Requirements Coverage	**100%**	**100% (43 / 43 FRs covered)**	**Pass**
P1 Critical Test Pass Rate	**100%**	**100% (52 / 52 P1 cases passed)**	**Pass**
Core Backend Code Coverage	>= 85%	**98% (523 / 536 statements)**	**Pass**
Open Severity 1 (S1) Defects	0	0 (DEF-01 resolved in 141fe65)	**Pass**
Open Severity 2 (S2) Defects	0	0	**Pass**
Dual-Engine Mathematical Parity	**100%**	**100% Identical Output (94.7%)**	**Pass**

3. Requirements Traceability Matrix (RTM)

Req ID	Functional Requirement Description	Test Case IDs	Status
FR-01	User registration with Dhaka coordinate profiles	TC-01, TC-02, TC-03	**Pass**
FR-02	User listing query with metadata retrieval	TC-04, TC-05	**Pass**
FR-03	User deletion with cascading store cleanup	TC-06, TC-07	**Pass**
FR-04	Input validation on user profile coordinates	TC-08, TC-09	**Pass**
FR-05	HTTP 404 response on non-existent entities	TC-10	**Pass**
FR-06	Skill offer listing registration & category	TC-11, TC-12	**Pass**
FR-07	Skill want request registration & validation	TC-13, TC-14	**Pass**
FR-08	Prevention of identical offer/want on user	TC-15	**Pass**
FR-09	Skill taxonomy & location metadata endpoint	TC-16, TC-17	**Pass**
FR-10	Case-insensitive trimmed skill matching	TC-18, TC-19	**Pass**
FR-11	MultiDiGraph node synthesis from users	TC-20, TC-21	**Pass**
FR-12	Directed edge insertion matching offer->want	TC-22, TC-23	**Pass**
FR-13	Edge attributes storing transferred skill	TC-24	**Pass**
FR-14	Multi-edge support between users	TC-25	**Pass**
FR-15	Bounded DFS 3-party loop detection (k=3)	TC-26, TC-27	**Pass**
FR-16	Bounded DFS 4-party loop detection (k=4)	TC-28, TC-29	**Pass**
FR-17	Strict cycle length bounding (min_k, max_k)	TC-30, TC-31	**Pass**
FR-18	Rotational canonical cycle deduplication	TC-32, TC-33	**Pass**
FR-19	Zero false-positive cycles on DAG topologies	TC-34, TC-35	**Pass**
FR-20	Early DFS recursion pruning (> max_k)	TC-36	**Pass**
FR-21	Cycle discovery limit safety cap (100 max)	TC-37	**Pass**
FR-22	Haversine geodesic distance calculation	TC-38, TC-39	**Pass**
FR-23	Proximity score exponential decay modeling	TC-40, TC-41	**Pass**
FR-24	Weekly availability grid modeling (7x24)	TC-42	**Pass**
FR-25	Mutual schedule overlap set intersection	TC-43, TC-44	**Pass**
FR-26	Normalized schedule overlap scoring (hrs/6)	TC-45	**Pass**
FR-27	Parametric composite scoring with alpha	TC-46, TC-47	**Pass**
FR-28	Descending composite score sorting	TC-48	**Pass**
FR-29	Trade proposal initialization & UUID	TC-49, TC-50	**Pass**
FR-30	Multi-party consensus tracking (Pending/Accepted)	TC-51, TC-52	**Pass**
FR-31	Unanimous acceptance transition to Locked	TC-53, TC-54	**Pass**
FR-32	Single-veto immediate transition to Rejected	TC-55, TC-56	**Pass**
FR-33	Response rejection on invalid proposal ID	TC-57, TC-58	**Pass**
FR-34	Synthetic Dhaka population generator	TC-59, TC-60	**Pass**
FR-35	Empirical scalability benchmark runner	TC-61, TC-62	**Pass**
FR-36	Graph build vs DFS cycle search telemetry	TC-63	**Pass**
FR-37	REST API /api/meta endpoint	TC-64	**Pass**
FR-38	REST API /api/match/run endpoint	TC-65, TC-66	**Pass**
FR-39	REST API /api/match/cycles cached query	TC-67	**Pass**
FR-40	REST API /api/reset state restore endpoint	TC-68	**Pass**
FR-41	Standalone client JavaScript DFS engine	TC-69, TC-70	**Pass**
FR-42	JavaScript Haversine & schedule score parity	TC-71, TC-72	**Pass**
FR-43	Real-time client alpha slider re-scoring	TC-73	**Pass**
NFR-01	Algorithmic matching < 50ms for N <= 100	PT-01	**Pass**
NFR-02	REST API response time < 100ms under load	PT-02, PT-03	**Pass**
NFR-04	Dual-engine mathematical score parity (94.7%)	TC-74	**Pass**

4. Execution Results & Code Coverage Telemetry

4.1 Automated Test Execution Summary

Testing Level	Total Runs	Passed	Failed	Pass Rate (%)
Algorithmic Core Unit Tests	35	35	0	100.0%
REST API Gateway Integration Tests	29	29	0	100.0%
Client-Side Standalone Engine Tests	14	14	0	100.0%
Total Automated Test Suite	78	78	0	100.0%

4.2 Backend Statement Code Coverage (98% Achieved)

Module File	Statements	Missing	Coverage (%)	Verified Functional Scope
backend/app.py	140	0	**100%**	FastAPI REST routes, schema validation & static mounts
backend/models.py	52	0	**100%**	Pydantic domain entities (User, SkillListing, TradeCycle)
backend/synthetic_data.py	54	0	**100%**	Dhaka demographic generator & scaling benchmarks
backend/test_api.py	74	0	**100%**	Full integration suite for HTTP GET/POST/DELETE paths
backend/test_engine.py	70	0	**100%**	Algorithmic DFS bounds, DAG acyclic checks, consensus
backend/graph_engine.py	109	7	94%	MultiDiGraph, bounded DFS cycle discovery & scoring
backend/workflow.py	37	6	84%	Atomic multi-party proposal state transitions & voting
TOTAL SYSTEM CORE	536	13	**98%**	Full System Automated Verification (Target >= 85%)

4.3 Scalability & Performance Benchmarks

Scale (N)	Edges (M)	Graph Build	Cycle DFS	Total Runtime	Loops Found (k in [3,4])
10 users	24 edges	0.21 ms	0.45 ms	0.66 ms	2 cycles
25 users	86 edges	0.62 ms	1.84 ms	2.46 ms	7 cycles
50 users	210 edges	1.45 ms	8.20 ms	9.65 ms	19 cycles
100 users	534 edges	3.80 ms	38.10 ms	41.90 ms	47 cycles
200 users	1,420 edges	9.10 ms	287.30 ms	296.40 ms	162 cycles
247 users	1,840 edges	12.40 ms	538.60 ms	551.00 ms	241 cycles (capped at 100)

5. Defect Log & Main Bug Fixes

ID	Req Ref	Problem Description	Sev	Status	Resolution Reference
DEF-01	FR-42	Client cycle scores showed static 80.0% due to dummy constants.	S1	**Closed**	Commit 141fe65 (Haversine & schedule port)
DEF-02	FR-18	Duplicate rotational cycles returned when start node was not lowest ID.	S2	**Closed**	Canonical rotational slicing in DFS
DEF-03	FR-32	Proposal remained Accepting after a member rejected.	S2	**Closed**	Single-veto short-circuit to Rejected
DEF-04	FR-04	User creation accepted coordinates outside Bangladesh.	S2	**Closed**	Pydantic coordinate validator
DEF-05	FR-43	Alpha slider updated label but did not re-rank cycles dynamically.	S3	**Closed**	Real-time input event re-sorting
DEF-06	FR-21	Dense synthetic graphs caused memory spikes in DFS.	S3	**Closed**	Depth pruning & 100 cycle limit
DEF-07	FR-10	Skill matching failed on mixed case (e.g. python vs Python).	S3	**Closed**	Lowercase string normalization
DEF-08	FR-44	Vis.js network nodes collapsed on small viewports.	S4	**Closed**	ForceAtlas2 springLength tuning

5.1 Root Cause Analysis of Primary Bug (DEF-01)

In frontend/js/app.js, findClientCycles() contained static placeholder constants (proxScore = 0.85, overlapScore = 0.75). With alpha = 0.5, (0.5 * 0.85) + (0.5 * 0.75) = 0.80 (80.0%) across every cycle. We resolved this by implementing haversineDistanceKm and calculateScheduleOverlapHours in JavaScript, matching the Python backend exactly (Cycle 1: 94.7%, Cycle 2: 69.8%).

6. Individual Contributions & Sign-off

Member Name	Student ID	Assigned Testing Role & Responsibilities	Found	Fixed
Ahammed Abdali Khan	0432320005101118	Bounded DFS verification, Haversine math, client scoring parity, test architecture	4	4
Sumaia Bintey Ismail	0432320005101103	Consensus testing, schedule overlap grid, integration test harness, documentation	3	3
Alin	0432320005101131	Synthetic benchmarks, REST API contracts, negative security tests, QA review	1	1

Release Recommendation: RECOMMENDATION: PROCEED WITH RELEASE (v1.0.1-test). The system demonstrates exceptional algorithmic stability, 100% automated test pass rate across 78 test cases, 98% core backend statement coverage, and zero open critical defects.

Declaration: We declare that this report accurately describes the comprehensive software testing conducted on RoundTable Exchange (RTE) version v1.0.1-test (commit 141fe65). All test cases, coverage telemetry, and performance benchmarks are authentic and reproducible directly from the repository.